Elliot Scher

(803) 873-3478 — ecscher@wpi.com — US Citizen
linkedin.com/in/elliotscher — github.com/ElliotScher

Education

Worcester Polytechnic Institute — Worcester, MA
B.S. Robotics Engineering — Expected May 2027

Technical Skills

Software Development: Java, Gradle, C/C++, CMake, Python, Git, Linux, Bash, Arduino, Raspberry Pi, MATLAB
CAD & 3D Printing: Onshape, Bambu Labs, Creality, Makerbot **Lab Equipment:** Soldering Iron, Multimeter, Oscilloscope, DC Power Supply

Experience

WPI Robotics Resource Center — Worcester, MA
XRP Curriculum Developer & Computer Vision Developer — Summer 2025

- Developed educational robotics curriculum for the open-source Experiential Robotics Platform (XRP).
- Designed and implemented a custom computer vision localization pipeline for FIRST Robotics Competition robots using fiducial detection and pose estimation.

Private Contract — Worcester, MA
Software Developer — Summer 2025

- Developed control software for a scientific instrument measuring optical density in agitated liquid samples.
- Implemented hardware interfacing and real-time data acquisition for automated laboratory measurements.

FIRST Headquarters — Manchester, NH
FRC Electrical Engineering Intern — Summer 2024

- Developed robot code templates for the 2025 FRC base robot in Java, C++, Python, and LabVIEW.
- Created a fiducial calibration utility integrated into the WPILib open-source robotics library.
- Collaborated with engineers and contributors supporting thousands of FRC teams worldwide.

STEM for Kids — Columbia, SC
Programming Instructor — Summer 2022

- Taught introductory programming concepts to K–5 students in classroom settings of up to 30 students.

Projects & Leadership

Robotic Arm Manipulation and Vision

- Developed inverse kinematics solutions for multi-joint robotic manipulators in MATLAB.
- Implemented cubic and quintic trajectory generation and motion planning for serial robotic arms.
- Built a computer vision pipeline for real-time colored object detection and target localization.
- Integrated perception and manipulation algorithms for autonomous object interaction tasks.

Autonomous Mobile Robot Navigation

- Implemented SLAM, odometry fusion, and path planning algorithms for autonomous robot navigation through a maze.
- Integrated LiDAR and onboard sensor data for real-time localization and obstacle avoidance.

Robotics Systems Software Platform

- Designed and implemented a modular robotics software framework inspired by Robot Operating System (ROS) architectures.
- Developed asynchronous publish-subscribe communication systems, finite state machines, and behavior tree control architectures.
- Implemented fault handling and recovery mechanisms for distributed robotic subsystems.
- Deployed autonomous navigation software on a simulated robotic platform for testing and validation.

Distributed Optical Density Measurement and Automation System

- Developed a distributed laboratory automation platform for monitoring optical density measurements in biological samples.
- Implemented a Raspberry Pi-based application layer for data visualization, experiment management, and system coordination.

- Designed embedded microcontroller software for real-time actuator and sensor control within the measurement system.
- Integrated serial UART communication between subsystems to support automated experimental workflows.

FRC Vision Localization System

- Developed a multi-camera vision localization system for FRC robots using AprilTag pose estimation.
- Implemented persistent Linux camera identification using udev rules and physical USB path mapping.
- Optimized low-latency pose publishing between coprocessor and roboRIO using NetworkTables.

GompeiLib Robotics Framework

- Developed a modular robotics software framework for FIRST Robotics Competition robots in Java.
- Implemented reusable subsystems for drivetrain control, vision integration, sensor management, and autonomous operation.
- Designed abstractions to simplify robot code development and improve maintainability across competition seasons.

WPICal Fiducial Calibration Tool

- Developed a fiducial calibration utility for the WPILib robotics software ecosystem used in FIRST Robotics Competition.
- Implemented camera calibration and AprilTag pose optimization tooling to improve robot localization accuracy.
- Contributed production-quality software integrated into the WPILib open-source library stack.

FRC KitBot Control Software

- Developed competition robot software for a FIRST Robotics Competition (FRC) KitBot platform in Java, C++, Python, and LabVIEW.
- Implemented modular subsystem architectures for drivetrain control and operator input handling.
- Designed reusable command-based software components to simplify testing, debugging, and future robot development.

FIRST Robotics Competition (FRC)
FRC Mentor, Team 190 (WPI, Mass Academy of Math & Science)

- Lead programming and controls mentor.
- Teach high school students Java, C++, and control theory for use on competitive robotics team.
- Manage Git organization and sub-team projects.
- Volunteer 50–60 hours/week during robotics season.

Team 190 Software Knowledge Base

- Developed a centralized knowledge base for FRC Team 190 to document best software development practices.
- Organized technical resources, guides, and onboarding materials to improve knowledge transfer between competition seasons.
- Maintained documentation for robot architecture, development workflows, tooling, and troubleshooting procedures to support student training and collaboration.

Autonomous Mobile Manipulator

- Developed an autonomous fruit harvesting and sorting mobile robot using VEX V5 Python.
- Implemented holonomic Mecanum X-drive kinematics with field-oriented control utilizing inertial gyro feedback.
- Built closed-loop proportional-derivative (PD) line-following control loops and sonar-based height adjustment.
- Designed an 11-state finite state machine (FSM) to coordinate vision tracking, claw collection, and sorting mechanisms.